Where machine learning for enantioselectivity on polysaccharide chiral stationary phases actually fails: a leave-one-paper-out reappraisal of 10,525 enantiomer pairs

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Abstract

Machine-learning models for enantioselectivity on polysaccharide chiral stationary phases increasingly report high accuracy, but almost always under random data splits. We re-examined this problem on CMRT, the largest public chiral chromatography dataset, curating it into 10,525 enantiomer pairs covering 530 DOI groups and 25 columns. Under a true leave-one-paper-out protocol, elution-order accuracy falls from 66.1 % under a random split to 52.9 %, indistinguishable from a within-paper permutation null (52.3 % ± 0.6). Between-paper variance accounts for 38.5 % of the total variance, but it cannot be predicted from compositional features, so domain adaptation is ineffective. The molecular signal is real within a single paper (median per-paper correlation r = +0.185) but falls to r = +0.0009 for an unseen paper, and adding training papers does not help. A variance decomposition of a single-laboratory grid shows that only 23.05 % of the variance is attributable to additive main effects of molecule and condition, the remaining 76.95 % being an unidentifiable molecule × condition interaction and noise. We therefore argue that the bottleneck lies not in model capacity or descriptor quality but in the data structure: adding condition metadata, building systematic molecule × condition grids, introducing replicate measurements and recording failed separations are the prerequisites for improving transferability.

Keywords: chiral chromatography; enantioselectivity; polysaccharide chiral stationary phase; machine learning; evaluation protocol; leave-one-paper-out; reproducibility; data structure

Highlights

- Under true leave-one-paper-out, elution-order accuracy on CMRT falls to 52.9 %, only slightly better than chance.
- Under the strict protocol no residual predictive ability is detectable: 52.9 % is not distinguishable from a within-paper permutation null (52.3 % ± 0.6).
- The 38.5 % between-paper variance is not compositional variance, and domain adaptation cannot recover it.

- The molecular signal is real within a single paper (r = +0.185) but does not transfer across papers (r = +0.0009).
- 76.95 % of the variance comes from an unidentifiable molecule × condition interaction and noise, not from insufficient model capacity; progress depends on data structure rather than data scale.

1 Introduction

Enantioselective separation on polysaccharide-based chiral stationary phases (CSPs) underpins the analytical quality control of chiral drugs, agrochemicals and flavor compounds [1]. Selecting the right CSP and mobile phase for a given analyte remains largely empirical, and the available combinations grow rapidly with molecular diversity. A model that could rank CSPs for a new analyte before any experiment would therefore have practical value, and a substantial literature has pursued this goal. The dominant approach builds a per-system quantitative structure–property model: a set of analytes measured on one (occasionally a few) CSP is described by molecular descriptors, and a regression is fitted to the selectivity factor α or to the elution order. Ensembles of steric and electrostatic chirality descriptors have been used for this purpose [2, 3], as have molecular graph representations [4] and classical two- and three-dimensional descriptor sets [5]; most recently, atom-centered scalar triple products of force-field geometries have been proposed for elution-order prediction [6]. These studies are typically built on a few tens of compounds — the largest same-protocol collections remain in the 43–48 analyte range — and are, by construction, tied to the selector on which they were fitted. Reviews and the authors themselves have flagged two natural limitations: the inability to transfer predictions to a different CSP, and the absence of an external test set [3]; the former makes predictions hard to detach from the column on which they were fitted, the latter leaves performance without independent testing.

Two developments make a broader assessment timely. First, the CMRT (Chromatographic Measurements of Retention Time) dataset [7] assembled 25,867 chromatographic records spanning 25 polysaccharide columns from about 640 literature entries, which for the first time makes cross-study modeling feasible; after curation and removal of exact duplicates these records collapse into 530 DOI (Digital Object Identifier) groups, one of which carries an unparseable DOI value in the source data and therefore represents 529 distinct publications. Second, reported elution-order accuracies reach 76–96 % [3], but this figure comes from a single 12-system study (6 CSPs × 2 mobile phases), not from a pooled cross-study result. At that level the question shifts from "can this be modeled at all" to "under what evaluation protocol does it hold, and does it transfer to another laboratory".

That second question is the subject of this work. Machine-learning evaluations on pooled chromatographic data face a specific hazard: measurements come from hundreds of laboratories, each with its own differences in instrument, column batch, mobile-phase preparation and calibration. If the data are split at random, records from the same paper appear on both sides of the split, and a model can exploit study-level regularities that carry no chemical information. Grouped or leave-one-study-out protocols (equivalent here to leave-one-paper-out, since one paper is one study) are standard practice in fields with pronounced batch structure [8, 9]; in chemically driven screening, strict splits that hold out unseen classes

are likewise used as a criterion for extrapolation ability [10]. Yet such protocols are not universal in this literature, and their effect on reported accuracy has — to our knowledge — not been quantified at scale on chiral chromatography data. We therefore take the CMRT collection, curate it into 10,525 enantiomer pairs with verified enantiomeric relationships (removing 597 candidate pairs that proved to be diastereomers), and re-examine the prediction problem under a true leave-one-paper-out protocol.

We report four findings. First, reported accuracy is almost entirely protocol-dependent: 66.1 % under a point-wise random split, 71.8 % under a grouped split, falling to 52.9 % under true leave-one-paper-out, a difference of up to 18.9 percentage points; a large part of the apparent performance of loose protocols comes from study-level leakage. Second, the between-paper variance accounts for 38.5 % of the total variance, but it is not compositional variance: it cannot be predicted from molecular or mobile-phase descriptors, and a planted-signal analysis excludes a composition-driven share of the between-paper variance of no less than 10 % at 95 % confidence, whereas the 5 % level is not excluded.

Third, the molecular signal is real but does not transfer: within one paper the median per-molecule correlation for new molecules is $r = +0.185$, whereas for an unseen paper it falls to $r = +0.0009$, and neither adding training papers nor improving the molecular description changes this. Fourth, the reason is structural: in the variance decomposition the interaction term and noise together account for 76.95 %, and the marginal and sequential shares must not be added (bootstrap intervals are given in Section 3.5); a model that is additive only in per-molecule and per-condition terms — which is what column-identity corrections and study-level domain adaptation are — is blind to that interaction term by construction, whereas a model that can represent interactions is not subject to this limit, and Section 3.5 gives the scope of that statement. This explains why descriptor enrichment, condition features and domain adaptation all fail for the same underlying reason: none of them adds the replication that would make the interaction identifiable. We close with a concrete and minimal prescription: the data structure needed to move past this ceiling depends on a handful of additional metadata fields and a change in study design, not on a larger dataset; we do not propose a stronger model, nor do we claim that more data or more expensive descriptors could raise this ceiling — rather, we set out to establish where it lies and what determines it.

2 Materials and methods

2.1 Data source and curation

We use the CMRT dataset [7]: an automatically extracted collection of chromatographic records for enantioseparation on polysaccharide CSPs, containing 25,867 records, 25 Daicel columns and about 640 source publications. CMRT provides, with each record, SMILES, retention time, flow rate, isopropanol (i-PrOH) volume fraction, column label and source DOI; these fields are the source of all descriptors and condition features below. The CMRT paper itself declares 25,847 molecules and 11,720 enantiomer pairs [7], whereas recomputation on the public data gives 25,866 records and 10,525 pairs [11]; the difference comes from the curation specification described below, in particular the 597 diastereomers rejected in enantiomeric-relationship verification. Each number has its own source: the former is the specification at

the time the source dataset was released, the latter is the specification at the end of the curation pipeline; the remainder of this paper uses the curated specification throughout.

Curation of this dataset is non-trivial. We performed nine processing steps in a fixed order (Table 1). Each step addresses a specific data problem: whether records are missing, whether retention times are physically plausible, whether DOIs are traceable, whether structures are parseable, where the dead time comes from, how replicate measurements are aggregated, and whether candidate pairs really are enantiomers. Several of these steps materially change downstream results. First, provenance: in about 5 % of DOI suffixes a hyphen was written as a slash, involving 1,307 records and 24 unique DOIs, and without repair the records cannot be traced back to the literature. Second, the physical quantities themselves: the dataset has no measured dead time, all 25 columns have the same 4.6 mm × 250 mm specification, the column volume is 4.155 cm³ and the pore volume is taken as 0.62 times the column volume, from which the dead time is estimated record by record and the retention factor is computed. The sensitivity of this choice is scanned systematically in Section 2.8.

Finally, measurement structure: replicates must be aggregated into condition units before pairing, otherwise α mixes the two mobile phases; the enantiomer-identity dimension is not redundant and cannot be omitted, and the four-tuple without it gives only 13,251 units. This step also fixes the unit definition on which all grouping and splitting below rests. After these treatments, 25,866 records collapse into 24,624 condition units, a further 597 candidate pairs prove to be diastereomers in enantiomeric-relationship verification, and 57 pairs are dropped because the retention factor is not positive. The complete curation pipeline is shown in Figure 6, and step-by-step details are given in Table 1.

Table 1 Data curation steps

Step	Treatment (from → to)	Rationale
1. Drop null values	25,867 → 25,867	SMILES, retention time, flow rate, i-PrOH and column have no missing values
2. Drop non-positive retention times	→ 25,866	Negative retention times are unphysical (1 record)
3. Repair DOIs	1,307 records / 24 unique DOIs rewritten	About 5 % of DOI suffixes have a hyphen written as a slash (e.g. 10.1038/s41467/019/13705/1); without repair the provenance chain is broken. The rule is to rewrite only the part after the first slash, and to require that exactly one slash remains after repair
4. Canonicalize structures	23,793 unique SMILES → 23,781 isomeric SMILES	Canonical isomeric SMILES generated with RDKit; zero parse failures
5. Estimate dead time and compute the retention factor	Dead time estimated per record as V_m/F	The dataset has no measured t_0. All 25 columns are 4.6 mm × 250 mm, so the column volume is 4.155 cm³ and the pore volume is taken as 0.62 times that; the sensitivity to this choice is scanned in Section 2.8 over $\varphi \in [0.55, 0.70]$
6. Fold into condition units	25,866 records → 24,624 units	A condition unit consists of molecule connectivity, column, i-PrOH fraction, flow rate and enantiomer identity. Replicate measurements must be aggregated before pairing, otherwise α

		mixes the two mobile phases. The enantiomer dimension is not redundant and cannot be omitted: the four-tuple without it gives only 13,251 units
7. Verify enantiomeric relationships	11,179 candidates → 10,582 true enantiomer pairs	597 candidates (5.3 %) proved to be diastereomers rather than enantiomers; see Section 2.2
8. Drop pairs with non-positive retention factor	−57 pairs	A retention time smaller than t_0 is unphysical (extraction error or gradient elution)
9. CIP orientation	→ 10,525 pairs	See Section 2.3

The curated dataset is summarized in Table 2. The main table contains 10,525 enantiomer pairs from 530 DOI groups, one of which is an unparseable value inherited from the source data and therefore corresponds to 529 real publications; these records cover 25 polysaccharide columns. The standard deviation of the target is 0.412, and the mean of the elution-order label is 50.7 %, close to balanced; the median number of rows per molecule is 1, i.e. most molecules were measured under a single condition. The several counts in the table correspond to different steps of the curation pipeline and cannot substitute for one another; the curated table is the basis for all splits in Section 2.6 and for the subsequent analyses.

Table 2 Summary of the curated dataset

Quantity	Value
Enantiomer pairs (main table)	10,525
Source publications (DOI groups)	530 (one group is an unparseable source value, therefore corresponding to 529 real publications)
Columns (polysaccharide CSPs)	25
Candidate pairs rejected as diastereomers	597 (5.3 % of 11,179 candidates)
Pairs dropped for non-positive retention factor	57
Standard deviation of the target y	0.412
Mean elution-order label (EMO)	50.7 %
Median rows per molecule	1

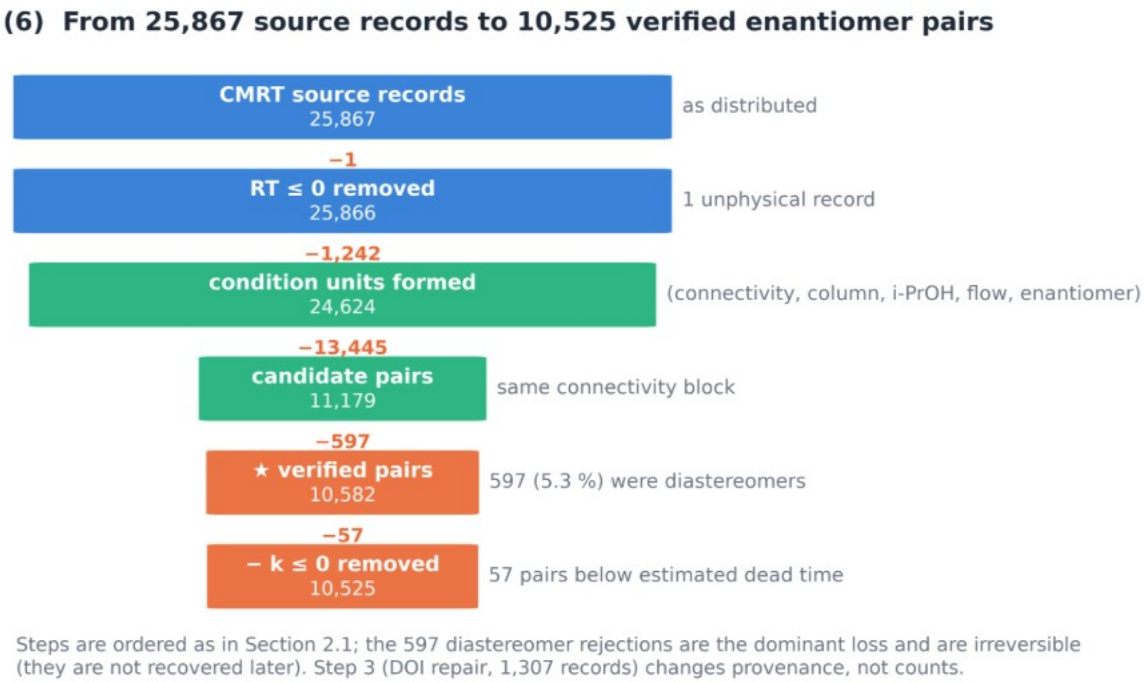


Figure 6 From 25,867 source records to 10,525 verified enantiomer pairs. Each bar corresponds to one step of the curation workflow of Section 2.1 and is drawn to scale on the record count, its width corresponding to the number of records output by that step, with the output count and the loss relative to the preceding step labeled. The dominant loss is the 597 pairs (5.3 %) judged to be diastereomers, which is irreversible. The DOI repair (1,307 records) and structure canonicalization change provenance and representation rather than counts, and are therefore described in the text rather than drawn as width changes.

2.2 Enantiomeric relationship verification

Sharing an InChIKey connectivity block does not establish that two structures are enantiomers; it identifies only the same skeleton, and diastereomers also fall within that skeleton. We therefore applied two mutually independent tests and required them to agree: M1 inverts all tetrahedral stereocenters and then compares canonical SMILES; M2 compares the /t layer of the InChI, requiring identical /t with different /m, or a character-wise inversion of /t. The two tests are independent in implementation — one compares stereochemistry directly, the other borrows the stereochemical layer of the InChI. They agreed completely on all 11,179 candidates with no disagreements, yielding 10,582 true enantiomer pairs and jointly rejecting 597 candidate pairs, 5.3 % of the candidate total. About 98 % of the rejected pairs differ at only one of two stereocenters, i.e. they are epimers. They are not enantiomers, there is no antisymmetric relationship between their ΔΔG and a pseudoscalar descriptor, and retaining them would contaminate the target; a pairing rule based only on the connectivity key would admit all of them. It should be noted that the InChI /m layer is not an oracle for R/S: it is normalized to each molecule's own canonical numbering and has no fixed correspondence to CIP labels, and in this data 73.7 % of molecules are m0=S and 26.3 % are m1=S. The /m layer is therefore meaningful only when comparing the two enantiomers of the same molecule, which is exactly how M2 uses it.

2.3 Target variable

The target of this paper is the CIP-oriented logarithmic selectivity factor, from which the binary elution-order label is obtained. Throughout we use EMO as the label for the elution-order prediction task and also as the name of its accuracy: EMO = 1 when the CIP-S enantiomer elutes later, and EMO accuracy is the fraction of pairs whose elution order is correctly predicted. Both the label and the accuracy are reported alongside a per-study majority-class baseline, because class balance differs greatly between studies. In this project EMO has always been used only as a label name; this paper defines its meaning and does not expand the acronym. The retention factor is computed according to the chromatographic definition, and the selectivity factor is the ratio of the retention factors of the two enantiomers, not the ratio of their retention times.

$$y = \ln \alpha = \ln\left[\frac{k(S)}{k(R)}\right], \qquad \mathrm{EMO} = 1\{y > 0\}$$

$$k = \frac{\mathrm{RT} - t_0}{t_0}, \qquad \alpha = \frac{k_2}{k_1}$$

R/S assignment uses the current CIP implementation of RDKit, not the elution order; 99.9 % of the pairs (10,516/10,525) are single-stereocenter, so k(S) and k(R) are literally the CIP-S and CIP-R retention factors. There are three reasons for CIP orientation. First, $\alpha \geq 1$ discards the elution order by definition, and the elution order is the most practically valuable output of a chiral separation; second, a pseudoscalar descriptor changes sign strictly between enantiomers, and an unsigned target cannot be aligned with it; third, an unsigned target makes "R elutes last" indistinguishable from "not separated". As an empirical check that CIP orientation is not systematically biased, the mean of EMO is 50.7 % and the standard deviation of the target is 0.412. We deliberately do not use ΔRT as the target: it inherits all the condition dependence of t_0 (flow rate and column geometry), whereas ln α is the quantity decoupled from conditions.

2.4 External datasets

We use two separately collected, protocol-uniform self-measured datasets to test external transferability; their measurement conditions differ from the literature data in CMRT and therefore constitute an external test of the models. DG25 [3] covers 42–43 compounds, 6 columns and 2 mobile phases, all measured in a single laboratory, and its 12-condition grid is central to the analyses of Sections 3.4–3.5; the target of this dataset is taken from the experimental enantioselectivity factor column of the source workbook, and the adjacent predicted column is discarded. PE25 [2] consists of measurements of 43 compounds under water and acetonitrile (ACN) conditions. We verified that the two datasets share 42 of the 43 compounds, so they are not two independent external sets and cannot be reported as two. DG25 serves as the primary external set, and PE25 is reported only in Supplementary Information S7-L: after removing analytes overlapping CMRT, n = 78, $R^2 = -0.173$, and EMO is 48.7 % against a 69.2 % majority-class baseline. PE25 is explicitly labeled as a near-duplicate of the same molecules under water and ACN conditions rather than as independent corroboration; four further analytes overlapping CMRT (Ibuprofen, mandelic acid, modafinil, rolipram) are removed from all external evaluations. The OD-RH column of DG25 has the same selector

chemistry as the Cellulose-a family in CMRT, which is why it serves as the aligned-selector test in Section 3.6.

2.5 Models and feature sets

All analyses were performed in Python 3.14.3, relying mainly on RDKit 2026.03.6 [12], scikit-learn 1.9.0 [13], NumPy 2.4.6, pandas 3.0.3 and SciPy 1.18.0; all results are regenerated by scripts from the curated table, and the archiving of the analysis chain is described in Section 2.9. RDKit handles structure canonicalization and descriptor computation, and scikit-learn handles model fitting. We deliberately use simple, standard baseline models rather than tuning a bespoke architecture, because the claim of this paper concerns the structure of the data rather than model capacity. The default model is gradient-boosted trees (HistGradientBoostingRegressor, 200–300 iterations, learning rate 0.06, at least 20 samples per leaf), with the matching classifier for the elution-order task. A standardized Ridge with a fixed $\alpha = 10$ is also used wherever the training set is too small for the tree model to be estimable; a fixed α rather than a cross-validated α is used both because cross-validation over about 15 rows is itself unstable and because the two arms being compared must share model capacity for the comparison to be meaningful.

This setting has a material effect on the results. In the within-laboratory setting of Section 3.3, the training half of a paper contains only about 15 rows, and with at least 20 samples per leaf a gradient-boosted model cannot make a single split and collapses to predicting the mean, which would make the "same laboratory" arm look like zero skill by construction. All small-sample comparisons therefore use Ridge, and the tree model serves only as a secondary check on papers large enough to support it. Similarly, R^2 is unusable when the variance of a test split is close to zero (values as low as -1×10^{15} were observed), and there the scale-free per-paper Pearson r is used instead.

On the feature side, three feature families are always used with the same splits and controls. L0 is a 28-dimensional baseline consisting of 25 RDKit two-dimensional molecular descriptors plus i-PrOH volume fraction, flow rate and estimated dead time, and serves as the reference throughout. The CSP descriptors are 167-dimensional, encoding selector backbone, substituent code and particle size, and are used for the extrapolation splits in which column identity is unavailable. The quantum-chemical chirality descriptors come in three generations: L1a-FF gives force-field geometry pseudoscalars (48 dimensions), L1-CF gives shape descriptors in a local chiral frame (89 dimensions), and L1b gives time-dependent density-functional theory (TD-DFT) electronic circular dichroism (ECD) rotational strengths (20 excited states); the definition and evaluation of the three generations are given in Section 3.7. Each feature family is evaluated under the same splits and controls, so that differences can be attributed to the representation itself. The geometries for L1a-FF and L1-CF come from ETKDG + MMFF94 (Merck Molecular Force Field) with a fixed random seed; L1b is taken from the CMCDS dataset [14].

2.6 Evaluation protocols

Seven splits are defined on the curated data table, progressively tightening study-level structure from point-wise random, through grouped and leave-a-batch-of-papers, to leave-one-paper-out; the split indices are released with the data for reproducibility. All seven splits use the same model and the same feature set,

with EMO as the primary comparison; S1 deliberately retains leakage to reproduce common practice, S3 and S3′ examine extrapolation across columns and across selector combinations respectively, and S4 holds out Murcko scaffolds to examine chemical-class extrapolation [15]. The number of folds ranges from 2 to 530 across the splits, and the held-out unit changes together with the fold count. Specific accuracies under each protocol are given in Section 3.1, and the grouping basis, fold count and purpose are given in Table 3.

Table 3 The seven evaluation protocols

Protocol	Grouping	Folds	Purpose
S1 random	None (point-wise 80/20)	2	Common practice; deliberately leaky
S2 grouped	(molecule, column)	5	Removes same-compound-same-column leakage
S2b leave-a-batch-of-papers	DOI, 10 groups	10	About 53 papers held out per fold
S2c true leave-one-paper-out	DOI, one paper per fold	530	Primary protocol
S3 leave-one-column	Column	25	Cross-CSP extrapolation
S3′ factor extrapolation	Substituent code × backbone	6 tasks	Holds out an unseen selector combination
S4 leave-scaffold	Murcko scaffold	5	Chemical-class extrapolation

In this dataset one source publication is one DOI, i.e. one study, so "leave-one-paper-out" and "leave-one-study-out" are synonyms here, and this paper uses leave-one-paper-out throughout. S2 groups by molecule and column, removing same-compound-same-column duplication; S2b holds out DOIs in 10 groups, about 53 papers per fold, averaging the paper effect within each fold; S2c holds out only one paper per fold and is the primary protocol of this paper. The ten-fold leave-a-batch-of-papers and the per-paper leave-one-out differ in meaning: the former removes a batch of papers each time, the latter removes only one study each time. Duplicate groups are keyed by (molecule, column) to prevent the same measurement from appearing on both sides of any split.

Three different evaluation samples must be distinguished. The S2c split itself contains 530 folds, one DOI per fold, covering all 10,525 rows; the descriptor-regression arm requires a molecule to fall within the common coverage of L1-CF and L1a-FF, leaving 523 publications and 10,363 rows; the per-fold summary counts only folds with at least 5 rows, covering 428 folds and 10,193 rows. The three numbers refer to three different samples, of which the 530 folds refer to the folds of the split itself. The difference in samples also determines which numbers belong to different quantities: the EMO arm of S2c evaluates the whole table (10,525 rows, 530 folds) and its accuracy is 52.85 %; the R^2 and per-fold statistics obtained from the two samples of 523 publications and 10,363 rows, and of 428 folds and 10,193 rows, are built on samples different from the full-sample arm, so the two specifications differ and cannot be compared directly.

2.7 Metrics, negative controls and statistics

We report R^2 and EMO for every protocol, and additionally, for the strict protocols, per-fold statistics including within-fold bias, Pearson r, Spearman ρ, R^2, and the number of folds with $r > 0$. Bias measures the difference between the fold mean and the mean of the target, r and ρ measure within-fold ranking, and

R^2 is affected by both. We further report R^2_{off}, the R^2 obtained after an oracle removal of each fold's mean offset, and R^2_{aff}, the result of an additional oracle affine rescaling on top of that. The latter two are oracle quantities, not performance measures; they bound how much of the error is attributable to offset and how much to ranking. We also report the grouping variance share η^2, and use the following identity as an audit tool: if a model has zero within-fold ranking ability, the affine-corrected error reduces to the within-fold variance, so an oracle-corrected R^2 cannot be interpreted without η^2.

$$R^2_{\text{aff}} \equiv 1 - \frac{E_w[\,\text{var}_f\,]}{\text{var}\,(y)} \equiv \eta^2$$

A large R^2_{aff} may be entirely a gift of the grouping structure. On S2c, R^2_{aff} = 0.439 against η^2(paper) = 0.381, both computed on the same 428-fold, 10,193-row sample; the derivation of the three η^2 samples is given in Section 3.1. Every quantitative claim is accompanied by a shuffled-label control run through the identical pipeline: after permuting the labels, $|R^2| \leq 0.064$ and EMO is about 50.2 % on this data, whereas the three permutations of S2c give 50.07 / 49.97 / 50.47 % (recorded as 50.17 %), which must be distinguished from the 50.67 % global majority class. The implementation of this control has two key points. First, shuffling must break the pairing between prediction and target: reshuffling predictions and targets together leaves the pairing intact and reproduces the original metric exactly — a control that appears complete but in fact tests nothing — so only the targets are permuted and the predictions keep their original order. Second, a null result requires a power statement: for the offset-predictability test of Section 3.2, a synthetic signal of known variance share is planted and the share that the same pipeline can recover is measured (39–93 % depending on the share), which converts "we observed zero" into a bounded exclusion. For comparisons that need to judge whether residual ability is detectable, a permutation null distribution is used instead: molecule labels are permuted within each study and repeated 200 times, and the null distribution is constructed from the median per-paper correlation and the fraction of papers in which it is positive. For the elution-order comparisons, the same within-paper permutation is applied to the labels and the whole pipeline is retrained: 200 permutations per scheme on the S2b protocol, and 30 permutations on the 530-fold S2c protocol with every fold refitted each time.

2.8 Dead-time sensitivity

Because t_0 is estimated rather than measured, we scanned the porosity $\varphi \in \{0.55, 0.58, 0.62, 0.66, 0.70\}$ (baseline 0.62), recomputing t_0, the retention factor and the target at each value and re-running the primary S2c protocol. Over this range the target correlates with the baseline at $r \geq 0.994$, the elution-order label flips for no more than 0.01 % of pairs, and the S2c R^2 is negative at every φ (−0.047 to −0.069). This scan is based on a fixed row set that is not the sample used for descriptor coverage in Sections 3.1–3.3 (all rows with a finite target, only folds with at least 20 rows). Consequently its baseline R^2 (−0.0688 at $\varphi = 0.62$, $n \approx$ 10,500) is not comparable with the −0.107 reported elsewhere, and only the variation across φ is interpretable. This scan tests the robustness of the target definition to the choice of t_0: within the given range, the qualitative conclusions are insensitive to this choice. However, the magnitude of R^2 moves by about 0.02 across the range, comparable to the descriptor effect sizes this study set out to detect, so the

estimated t_0 limits the interpretation of absolute R^2 values on this dataset; measured dead times are recommended for future data collection.

2.9 Code and data availability

Supplementary Information S1 is the supplementary version of the project's data-collection checklist; it lists the seven metadata fields and three measurement-structure requirements that should be recorded, and includes a ready-to-use collection template. The curated data table, the split indices, and all analysis and figure-generation code are archived and released with the manuscript, with the data and code hosted in a public archive repository, as detailed in the Declarations; these materials are sufficient to reproduce all numbers and figures in the main text. Of these, S1 addresses future data collection, whereas the archiving of data and code addresses reproduction of the results of this paper. The upstream CMRT dataset is publicly available (Zenodo record 10.5281/zenodo.7623903) [11].

3 Results

3.1 Reported accuracy depends almost entirely on the evaluation protocol

The accuracy of the same model on the same feature set differs across evaluation protocols by almost the whole range of values reported in the literature. Across the seven protocols (Table 3; Figure 1a plots only six of them), elution-order accuracy (EMO) is highest under splits that keep study-level structure intact and falls sharply once whole papers are held out: 66.13 % for the random split S1, 71.79 % for S2 grouped by the (molecule, column) combination, 56.46 % for leave-one-column-out S3, 58.71 % for leave-one-scaffold-out S4, dropping to 53.94 % for leave-a-batch-of-papers S2b and 52.85 % for S2c, which holds out one paper per fold (Figure 1a). Protocols that hold out whole papers report markedly lower accuracy than protocols that preserve study-level structure.

All accuracies in this section carry 95 % confidence intervals from a bootstrap that resamples whole source publications (2,000 draws; papers, not rows, are the independent unit): EMO = **66.13 % [64.55, 67.82]** under the random split, **71.79 % [70.18, 73.52]** under the grouped split, **53.94 % [51.73, 56.18]** under leave-a-batch-of-papers (S2b) and **52.85 % [50.63, 55.15]** under true leave-one-paper-out (S2c). The paired drop is **18.95 pp [16.59, 21.31]** from the grouped split and **13.28 pp [10.98, 15.58]** from the random split; S2c and S2b are not distinguishable (**−1.09 pp, [−2.44, +0.21]**).

These numbers have to be read against their own baselines. The per-fold majority-class baseline, weighted by fold size, is 50.87 %, 51.00 %, 55.81 % and 52.85 % for S1 to S4 and 52.87 % for S2b; by S2c the same quantity becomes an oracle baseline, because a new paper's class balance cannot be known in advance, and its value is 76.55 %. There is also a 50.17 % shuffled-label control (the three permutations gave 50.07 %, 49.97 % and 50.47 % in turn; too few draws, so the control band itself is poorly determined) and the 50.67 % global majority class. The two baselines serve different purposes: throughout this paper we use the global majority class (50.67 %) as the reference, and the per-study oracle majority-class baseline (76.55 % under S2c) is reported only to quantify how much of the apparent difficulty comes from study-level class imbalance. Against the global baseline the residual predictive skill of S2c is +2.18 pp, and

against the oracle baseline it is −23.7 pp. The sequence is not monotone: the EMO of S2 (71.79 %) is higher than that of S1 (66.13 %). This difference does not come from duplicate removal: only 1.4 % of S1 test rows share a same-compound same-column combination with the training set; it comes from the fact that after S2 groups by that combination, 4.4 % of test rows share a molecule with the training set, and the molecule-level signal is real (Section 3.3). The two protocols are optimistic in different ways; only S2b and S2c remove study-level structure, and the informative contrast lies between the two classes of protocols, S1 to S4 on one side and S2b and S2c on the other.

Seen from the protocol definitions, leave-one-paper-out means exactly one paper per fold: the dataset contains 530 DOI groups, one of which is an unparseable value inherited from the source data, so that in material terms it corresponds to 529 real publications, with a median of 18 rows each. A 10-fold grouping would hold out about 53 papers per fold and average the paper effect away, whereas the between-paper offset has a standard deviation of 0.256 and the fold means differ by only ±0.047; such a ten-fold protocol is not leave-one-paper-out and reports optimistically. Accordingly we constructed a genuine 530-fold S2c, in which every DOI is its own fold, and use it throughout as the primary protocol, with the ten-fold variant recorded as S2b. The EMO arm of S2c is evaluated on the full table of 10,525 rows and 530 folds; the descriptor-regression arm evaluates only the 523 papers with chirality-descriptor coverage, totaling 10,363 rows; the per-fold summary covers 428 folds and 10,193 rows in total, folds with fewer than 5 rows being dropped. The three numbers describe three different samples, defined in Section 2.6.

Oracle recalibration is not predictive skill. Under S2c, granting the model each held-out paper's own mean and scale — an oracle affine correction — raises R^2 to 0.439, and this number should not be read as performance. When within-fold ranking ability is zero, the affine-corrected R^2 equals exactly the between-group variance share; in these data η^2(paper) = 0.381, computed on the same 428-fold, 10,193-row sample as 0.439. The same statistic is 0.389 on the descriptor-coverage sample (10,363 rows) and 0.3854 on the full table (10,525 rows): these are three different samples, not three estimates of one number. Against 0.439, genuine within-paper ranking contributes only +0.06; both quantities carry the same small-sample optimism, with two oracle parameters fitted inside each test fold whose median size is only about 18 to 21 rows, so only their difference is interpretable, not either value on its own. On the same L0 feature set, R^2 falls from +0.275 under the random split S1 to −0.107 under true leave-one-paper-out S2c, a shift of −0.38; under S2c both arms have negative R^2, and adding the quantum-chemical chirality descriptors gives −0.048 (Figure 1b). Under S2c the oracle per-paper majority-class baseline for EMO is 76.55 %, close to the 76–96 % elution-order accuracies reported in the literature [3]. Different datasets, splits and models make this at most a coincidence that cannot be interpreted, but it does argue that elution-order accuracy should be reported together with a per-paper majority baseline.

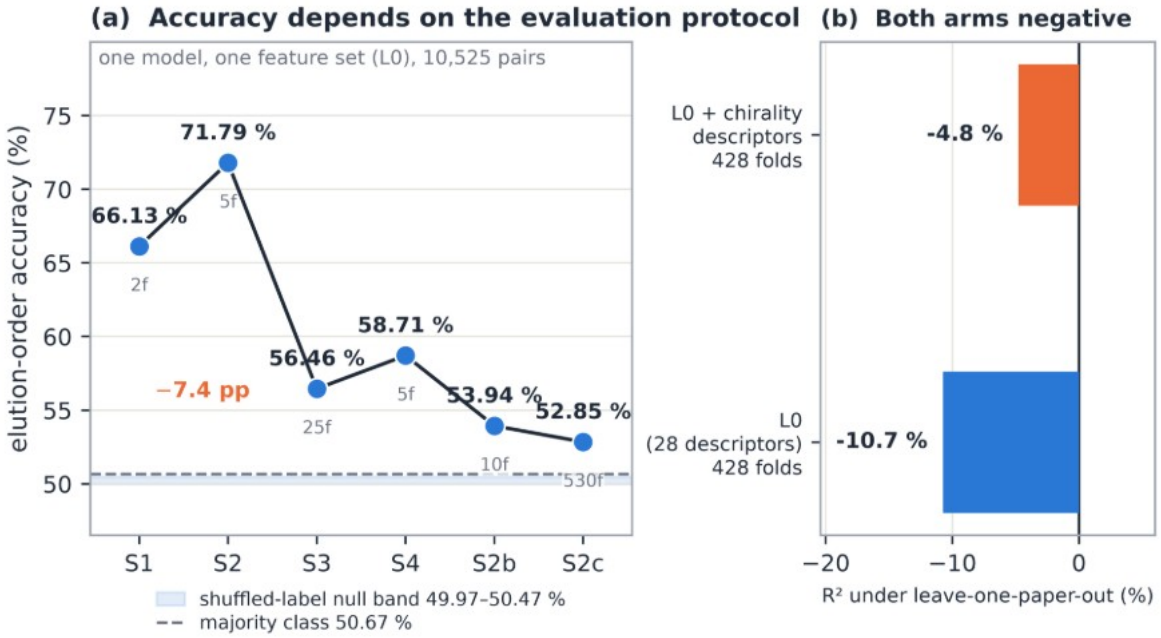


Figure 1 Protocol sensitivity of reported accuracy. **(a)** Elution-order accuracy under six protocols: S1, point-wise random 80/20; S2, grouped by (molecule, column); S3, leave-one-column (25 folds); S4, leave-Murcko-scaffold; S2b, leave-a-batch-of-papers (10 folds, about 53 papers per fold); S2c, true leave-one-paper-out (530 folds, one DOI per fold; 523 of the 530 papers retained at least 1 row after intersecting with the chirality-descriptor coverage). The shaded band is the range of three shuffled-label permutations (50.07 / 49.97 / 50.47 %; only three permutations, so the range itself is poorly determined), and the dashed line is the 50.67 % global majority class. **(b)** R^2 under S2c for the 2D baseline (L0) and for the arm augmented with quantum-chemical chirality descriptors, both negative; scoring uses the subsample with ≥5 rows per fold (428 folds, 10,193 rows), so the panel reports 428 folds rather than 530.

3.2 The between-paper variance is not covariate shift

Whether the between-paper offsets can be explained by observable composition differences determines whether they can be removed. Under the true leave-one-paper-out protocol, 38.5 % of the variance of the target $y = \ln \alpha$ lies between papers (η^2(paper) = 0.3854, row-weighted over the full 10,525-row table; the between-paper offset has a standard deviation of 0.256 against a total standard deviation of 0.412, Figure 2). If the offsets came from composition — papers studying different molecules or mobile-phase compositions — they should be predictable from each paper's own feature means, and domain adaptation could recover them. A direct test: for each paper we averaged its 25 RDKit 2D molecular descriptors together with its i-PrOH fraction, flow rate and estimated dead time, and predicted that paper's mean under leave-one-paper-out. $R^2 = -0.016$, with a random-label control of −0.004; using only the three condition features gives −0.009 and using only the molecular features −0.011, and no arm beats the null distribution.

A null result is only interpretable with a power statement, so we planted a synthetic composition-driven offset of known share (of the between-paper variance) and asked the identical pipeline to recover it; with five independent noise draws, the uncertainty is ±1 standard deviation of the recovery mean across draws. It recovered 1.95 % ± 1.68 % from a planted 5 % share, 7.37 % ± 0.97 % from 10 %, and 36.4 % ± 1.6 % from 39 %, a recovery rate of 39–93 % that rises with the planted share. The exclusion test uses a conservative bound, the recovery mean minus 2.13 times the draw-to-draw standard deviation (t_4 = 2.13, a one-sided 95 % bound; taking the bound on the standard error of the mean would be 2.2 times narrower): shares of at least 10 % of the between-paper variance are excluded, with a lower bound of +5.3 % above

the observed value of −1.58 %; the 5 % level is not excluded, its lower bound of −1.62 % lying below the observed value. The between-paper share of the total variance is 38.5 %, so the excluded 10 % corresponds to about 3.9 % of the total variance, and the observed value of −1.58 % lies below every planted point (Figure 2). The between-paper variance is therefore not composition in disguise, but this claim has a boundary: the composition features deliberately exclude stationary-phase identity, so the correct statement is that the offsets are unpredictable given molecular and mobile-phase descriptors, not that nothing about them is knowable; Section 3.4 will show that the selector in fact matters a great deal, but as a reordering rather than an overall offset.

Another model-free probe uses the data in which the same molecule on the same column was measured in two or more papers (124 groups, 274 rows; the looser match of same molecule and same selector family gives 187 groups, 405 rows). Within these groups the within-group standard deviation is 0.338 against a total standard deviation of 0.410 for the subset, so the within-to-total standard-deviation ratio is 0.82 (the between-group standard deviation is 0.324, a between-to-total ratio of 0.79; the two ratios are descriptive summaries computed with different degrees of freedom and do not combine as a variance decomposition). The ceiling implied by the within-to-total ratio is $1 - 0.82^2$, about 0.32. Converting the 95 % interval of the within-to-total ratio into an R^2 interval gives [−0.67, +0.36], and at this sample size the point estimate cannot be distinguished from zero. Separately, one historical column name in the source data, although written as a within-paper standard deviation, actually corresponds to the within-group standard deviation; this paper uses it in the latter sense.

The broader match of same scaffold plus same selector family (3,655 rows) gives a point ceiling of −0.75 and is uninformative by construction (it admits different substituents); it is reported only to show the direction of the bias. This subset is selected and small, accounting for 2.6–3.8 % of the full table, and we do not extrapolate from it. Within-paper replication is sparse rather than absent: there are 609 within-paper replicate groups (same molecule, same column, same mobile phase, same flow rate and same enantiomer under the same DOI, 1,268 rows in total), of which 357 groups do have different retention times; group sizes are mostly 2 (587 groups), but 22 groups hold 3 to 7 records, and 3.8 % of pairs carry a non-zero reported relative standard deviation of the retention factor, with maxima of 0.95 and 0.77 for the two enantiomers. In the cleaned table a cell of one molecule and condition holds at most eight replicate records, which is still not enough to separate hard-to-predict from a noisy target itself; this is a limitation of coverage, not a complete absence.

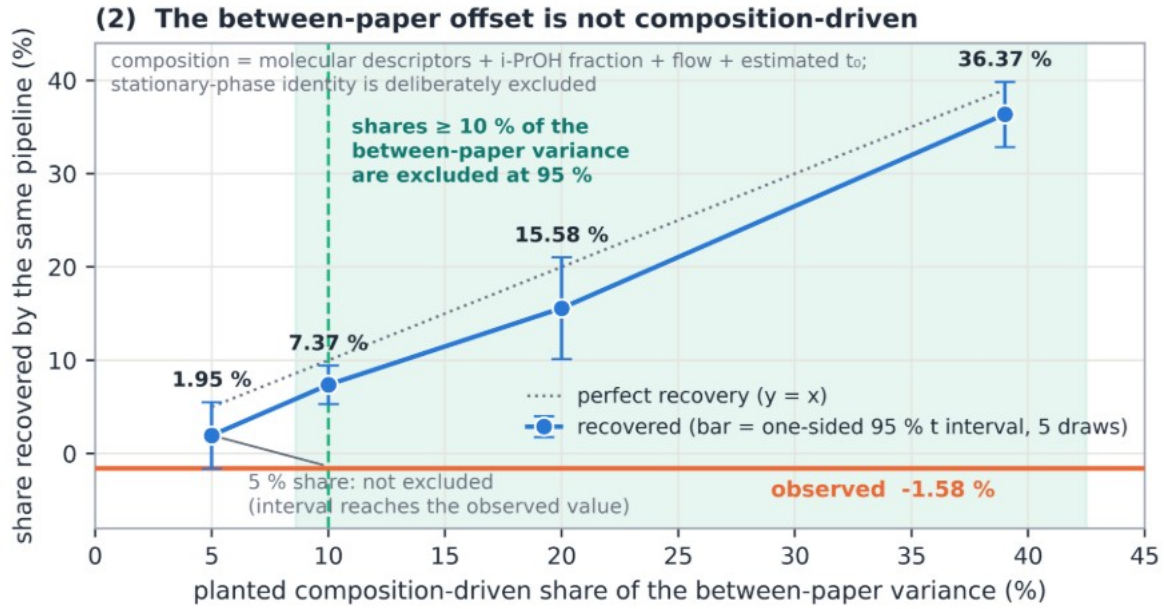


Figure 2 Predictability of the between-paper offset, with a power analysis. The orange line is the observed value (−1.58 %) obtained when a paper's mean target is predicted under leave-one-paper-out from its own mean composition alone (molecular descriptors, i-PrOH fraction, flow rate and estimated dead time); the blue line is the recovery of the same pipeline after planting a composition-driven offset of known share (of the between-paper variance). Error bars are $\pm t_4$ × the draw-to-draw standard deviation of the recovery over five independent draws ($t_4 = 2.132$) and delimit the draw-to-draw standard deviation rather than the standard error of the mean—the latter would be 2.2 times narrower with five draws and would flip the verdict at the 5 % level. With the conservative criterion, planted shares of ≥10 % are excluded while the 5 % level is not, which is precisely the claim made in this paper. Both axes are in percent; the composition features deliberately exclude stationary-phase identity, so the result constrains only offsets given the molecular and mobile-phase descriptors.

3.3 The molecule-level signal is real but does not cross laboratories

The results above could be read as saying that there is no learnable signal, but that reading is wrong, and the distinction matters. We compared two settings with the same model class (standardized Ridge with a fixed regularization strength, regularization parameter 10) and the same 28 features. Within one laboratory, for each paper with at least 25 rows we split its molecules 50/50, grouping the split by molecule so that a compound measured on several columns never straddles the split; one half is used for training and the other half for prediction. Across 170 papers the median per-paper Pearson r is +0.185, and 80 % of papers are positive.

Two features of the within-paper protocol could in principle manufacture this correlation: the condition features (i-PrOH fraction, flow rate and dead time) are shared between the training and test halves of a paper, and t0 enters both the target and the feature set. Repeating the identical 170-paper, eight-replicate split with the feature families swapped resolves both concerns. With molecular descriptors alone the median per-paper r is **+0.169** (79 % of papers positive); with condition features alone **+0.110** (61 %); with both **+0.185** (80 %). Adding the condition features to the molecular description leaves the per-paper correlation essentially unchanged: the median of the 170 paired differences is **−0.0001** (Wilcoxon signed-rank p = 0.41), even though the two-arm medians themselves differ (+0.169 versus +0.185). Removing t_0 alone from the full set changes the median by **+0.0006**. The within-study signal is therefore carried by the molecular description, not by condition information shared across the split, and it is not an artefact of the dead time used to construct the target.

In the across-laboratory setting, training on other papers and predicting a held-out paper gives a median per-paper r of +0.0009, close to a coin flip (49 % of papers positive). The two settings share the same metric (per-paper r, which is scale-free), so their medians are comparable; their test-set sizes are not comparable, the within-study half holding about 15 rows and a whole held-out study 1 to 120 rows, so we compare medians rather than per-paper values. The contrast lies far outside a permutation null: permuting molecular labels within each study 200 times gives a null median per-paper r of −0.012 ± 0.014 and a null proportion of positive papers of 47 % ± 3.6 %, against the observed +0.185 and 80 % (Figure 3). A sweep over five random seeds shows that this is not a split artefact: the within-laboratory median stays in [+0.176, +0.200] (77–82 % of papers positive) and the across-laboratory median stays in [−0.042, +0.011].

Adding data does not close the gap. Sweeping the number of training papers k from 1, 2, 5, 10, 20, 50, 100 to 250 and evaluating on unseen papers, the pooled R^2 remains negative throughout (k = 250 gives −0.0498); even when an oracle affine rescaling of the predictions is allowed, forgiving any scale miscalibration, the recovered R^2 never exceeds 0.03 at any k, and the curve is flat over a 250-fold increase in training data. The scope of this arm needs to be stated: the within-laboratory training half contains only about 15 rows, so +0.185 measures what is achievable at that sample size and is not an upper bound, and it should rise with more within-laboratory data; this arm is also restricted to the 170 of the 530 studies with at least 25 rows, so it describes what is achievable for studies of that size, not for the collection as a whole. By contrast, the flatness of the across-laboratory curve does not depend on the training-set size of any single fold.

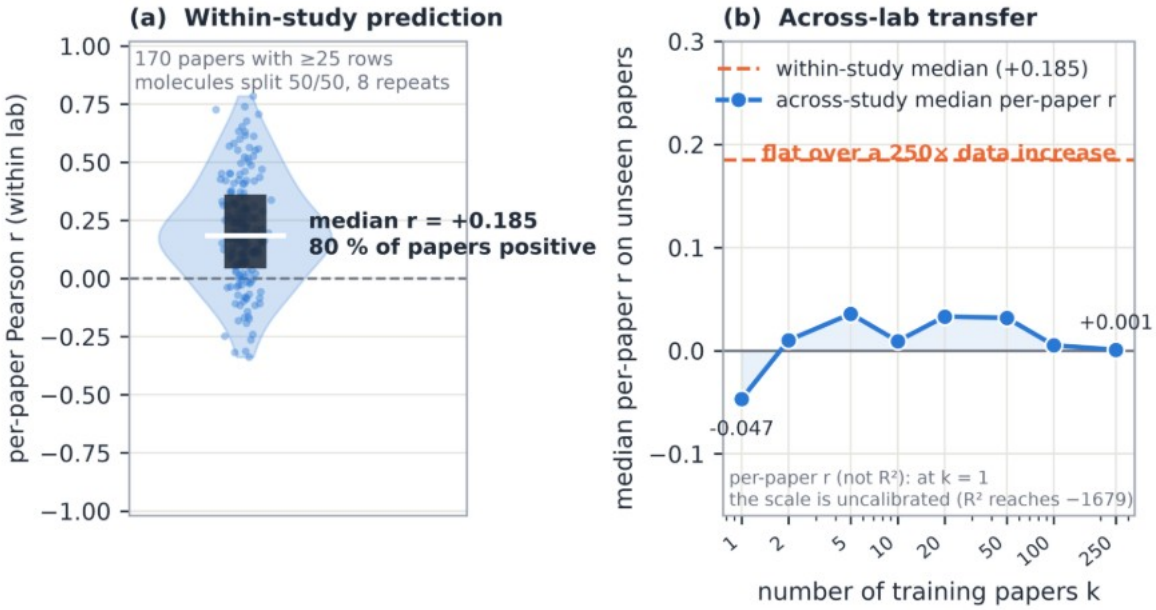


Figure 3 Within-laboratory versus across-laboratory prediction. **(a)** Distribution of per-paper Pearson r for the 170 papers with ≥25 rows (of 530), each split 50/50 by molecule (8 repeats, averaged by group), with a median of +0.185 and 80 % of papers positive; a 200-draw within-study permutation null gives −0.012 ± 0.014. **(b)** Across-laboratory learning curve: the median per-paper r when k papers are used for training and the remainder for testing, with the within-laboratory median marked by the orange line. Because the prediction scale is uncalibrated at k = 1 (R^2 down to −1679), the scale-free per-paper r is used here. Over five random seeds, the within-laboratory median stays in [+0.176, +0.200] and the across-laboratory median in [−0.042, +0.011].

3.4 The failure is a reordering, not an offset — a model-free demonstration

A natural objection to model-based comparisons is that the failure comes from the model itself; to exclude this possibility we measured reproducibility directly, without any model or descriptor. The De Gauquier dataset [3] measured the same 42 to 43 compounds under 12 conditions in a single laboratory, the 12 conditions being 6 columns combined with 2 mobile phases, with column identity determined by its selector. Holding the compounds and the laboratory fixed, any disagreement between two conditions is attributable to the conditions alone. The 12 conditions were paired two at a time and grouped by the type of condition that changed; pairwise correlations are not independent — each of the 12 conditions participates in 11 pairs, and the same-compound requirement couples them further — so we report bootstrap intervals clustered by compound (2,000 resamples over the 43 compounds) together with a paired bootstrap for the between-group differences, with the same resampled compound set used for both groups of each contrast. The results are in Table 4.

Table 4 Ranking correlation between conditions

What changed	Pairs	Median r	95 % interval (compound-clustered)	Paired bootstrap difference vs both-changed (95 % interval)
Mobile phase only (same column)	6	+0.560	[+0.274, +0.829]	+0.500 [+0.215, +0.769]
Column only (same mobile phase)	30	+0.128	[+0.001, +0.250]	+0.061 [−0.059, +0.190]
Both changed	30	+0.084	[−0.044, +0.180]	—

Note: r is the median Pearson correlation between pairs of conditions; intervals are 95 % bootstrap intervals clustered by compound; the last column gives the paired bootstrap difference from "both changed" and its 95 % interval.

When the column is fixed and only the mobile phase changed, ranking is preserved significantly better than when both are changed: the median r is +0.560, 95 % interval [+0.274, +0.829] (6 pairs), and the paired bootstrap difference is +0.500, interval [+0.215, +0.769], $P(\Delta \leq 0) = 0.0005$. When only the column is changed the median r is +0.128, interval [+0.001, +0.250] (30 pairs); column-only and both-changed are not distinguishable, the difference being +0.061, interval [−0.059, +0.190], $P(\Delta \leq 0) = 0.17$, so the arm that can be resolved is the mobile-phase one; the column-only arm has a median of +0.128, slightly above zero but with a 95 % interval that touches zero. The effect tracks chemical similarity: cellulose–cellulose column pairs transfer best, with r = +0.664 between LC-2 and LC-4 and a sign agreement rate of 82.6 %; amylose–amylose column pairs are intermediate, with +0.304 between LA-1 and LA-2; and amylose–cellulose column pairs transfer poorly or even inversely, with −0.291 between LA-2 and LC-4 (Figure 4b).

The decisive observation is that the selector's main effect on the target is extremely small: within this grid η^2(column) = 1.0 % and η^2(condition) = 1.4 %, computed on the corrected 494-cell sample used in Section 3.5; the uncorrected 549-row coding gives 1.3 % and 1.5 %, and CMRT gives a consistent η^2(column) = 2.6 %. A main effect of about 1 % is precisely what domain adaptation or column-identity correction can absorb, yet a change with such a small mean effect can still drive the cross-condition ranking

correlation toward zero. A small mean effect and an almost complete loss of ranking can coexist, and it is the latter that a predictive model must survive. The boundary of this demonstration comes from the size of the grid: each condition pair rests on only 42 to 43 compounds, each compound–condition combination has exactly one observation, so measurement noise cannot be estimated; the comparisons are between conditions, and we do not claim that r = 0.560 represents a noise ceiling.

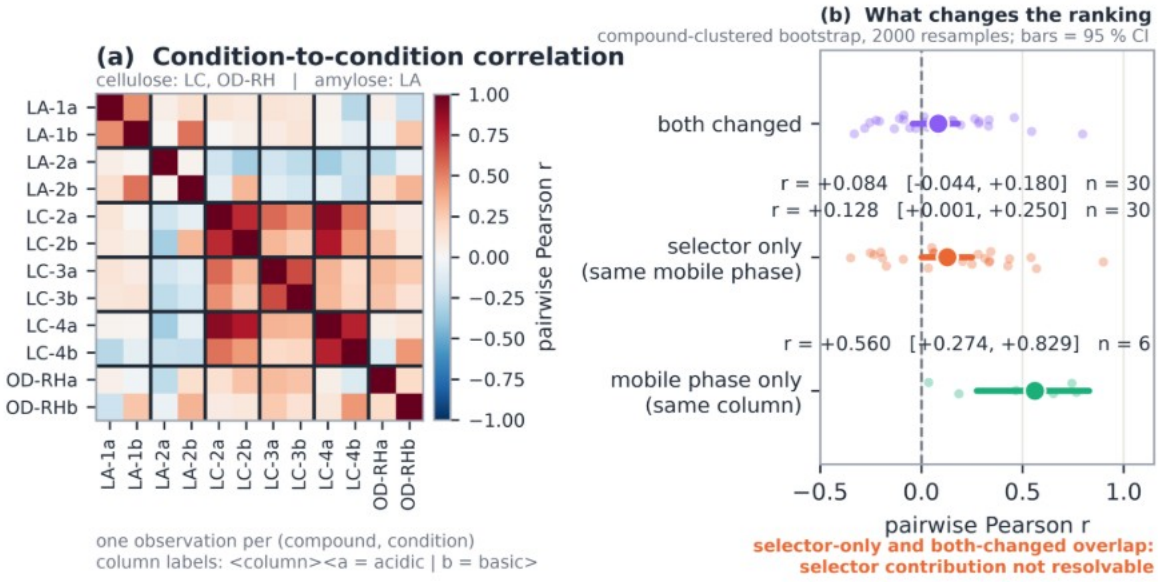


Figure 4 Condition-to-condition reproducibility, without any model or descriptor. The data are the De Gauquier grid: 42–43 compounds × 6 columns × 2 mobile phases, measured in one laboratory. **(a)** Correlation matrix between all 12 conditions, labeled by column name plus mobile phase (a acidic, b basic), with separation lines marking the cellulose (LC, OD-RH) and amylose (LA) blocks. **(b)** Median pairwise r for each class of condition change, with error bars showing the compound-clustered bootstrap 95 % intervals (2,000 resamples of the 43 compounds). Changing only the mobile phase largely preserves the ranking (+0.560 [+0.274, +0.829]); the selector-only (+0.128 [+0.001, +0.250]) and both-changed (+0.084 [−0.044, +0.180]) medians are not distinguishable from each other (paired bootstrap +0.061, 95 % interval [−0.059, +0.190]). Each (compound, condition) cell holds only one observation, so measurement noise cannot be estimated.

3.5 An additive description of the label covers about a quarter of the variance

Sections 3.2–3.4 point to a structural rather than a modeling explanation, and it can be made quantitative by writing the target as

$$y(m,c) = \mu + a_m + b_c + h(m,c) + \varepsilon$$

where a_m is the molecule main effect, b_c the condition main effect, h(m, c) the interaction and ε the noise. A predictor that is additive in per-molecule and per-condition terms is blind to h by construction; this class includes column-identity corrections and study-level domain-adaptation offsets. Descriptor-based machine learning is not restricted in this way: a gradient-boosted tree over concatenated molecular and condition features can in principle represent h, and Section 3.7 uses exactly such a model. The variance share here is therefore strictly the oracle main-effects R^2 of the label — the share of variance that any label-additive rule can reach when both main effects are known — and not a bound on arbitrary machine learning.

Accordingly we do not compare this variance share with reported accuracy percentages: the two are different quantities and are not monotonically related here. Under the grouped protocol of Section 3.1, R^2 = 0.27 coexists with 71.8 % EMO; on the separated part of the grid, an additive rule reaches 72 % EMO (Supplementary Information S7-K). The De Gauquier grid is 95.7 % complete and unbalanced: 43 resolvable molecules and 12 conditions give 516 theoretical cells, of which 494 are populated, and only 30 of the 43 molecules span all 12 conditions, so the decomposition is a two-way least squares on populated cells only. The 55 records with missing molecular connectivity are dropped rather than assigned to a level; encoding them as all-zero rows (that is, as the reference molecule level) would count unstructured molecules as real ones. Shares are reported sequentially, with the molecule entering first, so that the molecule and condition terms sum exactly to the additive share; the single-factor marginal shares (21.8 % for molecule, 1.4 % for condition) cannot be added to them. The 95 % interval is a bootstrap over molecules (200 resamples, fixed seed) computed on the same 494 rows. Three sensitivity analyses — 549 rows as 48 levels giving 22.33 %, the balanced 30-molecule sub-grid (360 rows) giving 20.18 %, and the legacy reference-level coding giving 22.29 % — leave the conclusion unchanged; grid completeness is 95.74 %. The sequential decomposition is summarized in Table 5.

Table 5 Additive variance decomposition

Component (sequential)	Share of variance
Molecule main effect	21.80 %
Condition main effect (sequential)	1.25 %
Oracle main-effects R^2 (additive description)	23.05 %
Molecule × condition interaction plus noise	76.95 %

Note: shares are reported sequentially, with the molecule entering first, so that the molecule and condition terms sum exactly to the oracle main-effects R^2; the interaction term and noise are not separable.

Because the 12 conditions are themselves a sample from the space of conditions, we report two bootstrap intervals: resampling molecules only, and resampling molecules and conditions jointly. The main-effects R^2 is 23.05 % (Figure 5); a non-parametric bootstrap that resamples molecules only (200 draws, seed 20260917) gives a distribution with mean 24.9 % and a 95 % percentile interval of [17.5 %, 32.9 %], joint resampling widens the interval to [18.2 %, 40.8 %], and an independent implementation with a different joint-resampling scheme gives [16.6 %, 50.7 %]. The point estimate lies at the 32nd to 40th percentile of the molecule-only bootstrap, so both intervals are distributional summaries rather than centered confidence intervals, and the conclusion is anchored on the weaker statement that holds under every scheme: the main-effects R^2 stays below 50 %.

The shares are a sequential decomposition and are not additive beyond the molecule and condition pair; the marginal condition share of 1.37 % cannot be added to the other terms. The total main-effects share is independent of entry order: molecule first gives 21.80 % plus 1.25 %, condition first gives 1.37 % plus 21.68 %, both 23.05 %, so the ordering only determines the split between the two factors and does not change the additive total. Three sensitivity analyses bound the sample choice, 22.29 %, 22.33 % and 20.18 % corresponding to an operational band of 20.2 to 23.1 %, and the direction of the conclusion is unchanged. This unifies the negative results obtained along three routes that should be unrelated: enriching

the molecular descriptors improves only a_m, adding condition features or domain adaptation adjusts only b_c, and an oracle affine rescaling changes only the scale, none of them reaching h; more precisely, they fail because h is not identifiable under this data structure, not because no model can represent an interaction. It also explains the pattern of Section 3.4: changing the condition reorders the molecules, which is exactly the statement that h is not equal to zero.

The separated cells of the same grid ($\alpha \neq 1$, 215 cells) provide a direct comparison: majority-class baseline 58.6 %, condition main effect 58.1 %, oracle two-way additive 72.1 %, leave-one-molecule-out with the condition known 76.3 %, and per-molecule oracle majority class 77.7 %, with an area under the receiver operating characteristic curve (AUC) of 0.76 to 0.84 (Supplementary Information S7-K). The variance share is thus not an accuracy ceiling: inside the fully crossed grid, additive descriptions predict elution order well, and what fails is their transfer to new studies, because the conditions under which a new study measures its molecules are neither recorded nor replicated. The constructive implication is concrete: closing this gap requires data in which the same molecule is measured under several conditions with replication, not more molecules and not more expensive descriptors; that is precisely the structure the public data in this field lack.

The boundary of this decomposition comes first of all from the data: it comes from a single dataset (one laboratory, 43 molecules), is a demonstration of a principle rather than a universal constant, and should not be read as a bound for the 530 DOI groups of CMRT, whose molecular diversity would only make the molecule main effect smaller. Because each cell holds exactly one observation, the interaction and measurement noise are not separable, so 23.05 % is an upper bound on what a label-additive description can reach; a leave-one-molecule-out check on the same grid with the condition known, using real molecular descriptors, returns $R^2 = -0.27$, below the main-effects value, as expected when only about 40 molecules are available to estimate the molecule effect. The variance share and task accuracy are different quantities; nothing here rules out the predictive ability of additive models within a fully crossed grid, and what fails is their transfer to new studies.

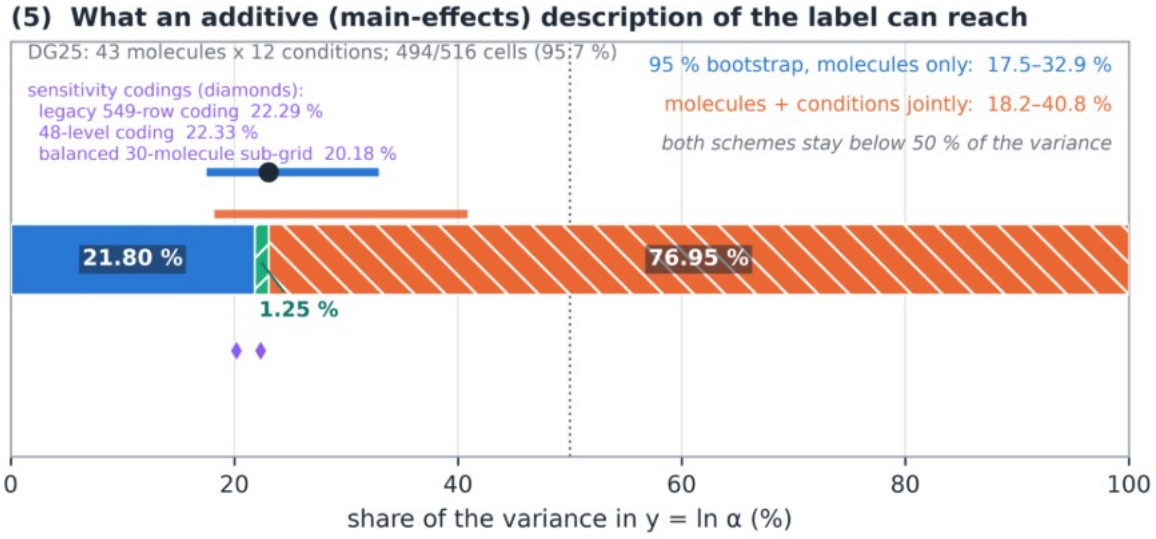


Figure 5 What an additive description of the label can reach. Two-way variance decomposition of the De Gauquier grid (43 resolvable molecules × 12 conditions): 95.7 % complete and unbalanced (494 of 516 theoretical cells populated, 30 of the 43

molecules spanning all 12 conditions, and the 55 rows whose molecule connectivity is missing excluded from the main analysis). The sequential decomposition gives a molecule main effect of 21.80 % and a condition main effect of 1.25 %, together making up the 23.05 % oracle main-effects R^2 of any label-additive rule, whose 95 % bootstrap intervals are [17.5 %, 32.9 %] (resampling molecules) and [18.2 %, 40.8 %] (resampling molecules and conditions jointly); these are sensitive to the scheme and are therefore presented only as distributional summaries. The remaining 76.95 % is the molecule × condition interaction plus noise, which is not identifiable when each cell holds only one observation; on the separated subset of this grid an additive rule reaches an elution-order accuracy of 72.1 %, showing that this variance share is not an accuracy ceiling. Three alternative codings give an operational band of 20.2–23.1 %.

3.6 External validation fails even when the selector is aligned

Internal splits may understate the reach of a model; external validation is the stronger test. This study uses the De Gauquier dataset [3], which measures 42 to 43 compounds on 6 columns under 2 mobile phases in one laboratory, a near-complete but unbalanced grid in which 494 of the 516 compound–condition cells are populated, that is 95.7 %, with the selector and the mobile phase varied systematically. We first chose the OD-RH column because its selector chemistry, cellulose tris(3,5-dimethylphenylcarbamate), is the same as that of CMRT's Cellulose-a family (columns OD, OD3, ODH, IB), which is the smallest domain gap available in public data. Training on CMRT Cellulose-a (2,738 rows) and predicting the 73 non-overlapping OD-RH measurements gives $R^2 = -0.802$; using all of CMRT gives $R^2 = -0.477$. Restricting the test set to compounds that are actually separated ($\alpha \neq 1$, $n = 30$) gives −0.601 for the aligned arm and −0.316 for the all-CMRT arm. A shuffled-target control (five repeats) gives -0.756 ± 0.157 for the aligned arm, that is, the real result is indistinguishable from chance and in fact slightly lower; both arms are below the constant-mean predictor ($R^2 = 0$).

Aligning the selector bought no transferability and in fact made things worse, because the aligned training set is 4 times smaller. To rule out the objection that OD-RH happens to be a bad choice, we repeated the test on all six columns (Figure 7). Every column returns a negative R^2 under all-CMRT training (−0.149 to −1.441); the cellulose columns are not systematically better than the amylose ones, with medians of −0.443 and −0.243 respectively; and the mean across columns of −0.531 does not beat the shuffled control of -0.465 ± 0.284. The six-column control uses only one shuffle per column, so ±0.284 is merely the between-column spread of the control rather than a permutation standard deviation, and we use it descriptively only. The predictions are also strongly shrunk: their standard deviation is 0.12 to 0.13 against a target standard deviation of 0.13 to 0.32. The interpretation of this failure is subject to two limitations. Each column contributes only 70 to 78 test measurements; the mobile-phase mode of the datasets also differs, CMRT being normal phase and De Gauquier water with acetonitrile, so the failure is a composite of laboratory and condition shift and cannot be attributed to the laboratory alone. Between 49 % and 72 % of the external samples are non-separated ($\alpha \equiv 1$), 58 % overall, and per column OD-RH 59 %, LC-2 49 %, LC-3 72 %, LC-4 53 %, LA-1 57 % and LA-2 58 %, against 0.16 % in CMRT, which depresses R^2 independently of transferability.

S3′ leaves out an entire selector family rather than a column; six extrapolation directions were run and **every one gives $R^2 < 0$**, from **−0.02 to −0.30** with the best self-built CSP description (**−0.10 to −0.66** with the vendor labels). Adding the self-built CSP structural descriptors changes R^2 by ≤0.01 in four of the six directions; the largest improvement (+0.50, still negative) occurs in the hardest direction (amylose → cellulose, subset f). The failure is therefore not a matter of selector description.

Column identity alone does not explain the collapse. Within a single column the literature-style random split reaches **74.61 %** EMO (per-column majority baseline **55.83 %**), whereas the same column under a leave-one-paper-out split reaches **52.73 %** against a per-column oracle majority baseline of **83.76 %**; the drop is negative in **24 of the 25 columns** and exactly zero in the remaining one (median **−20.5 pp**). Because every column is analysed separately, this failure cannot be a mixture of column and mobile-phase effects.

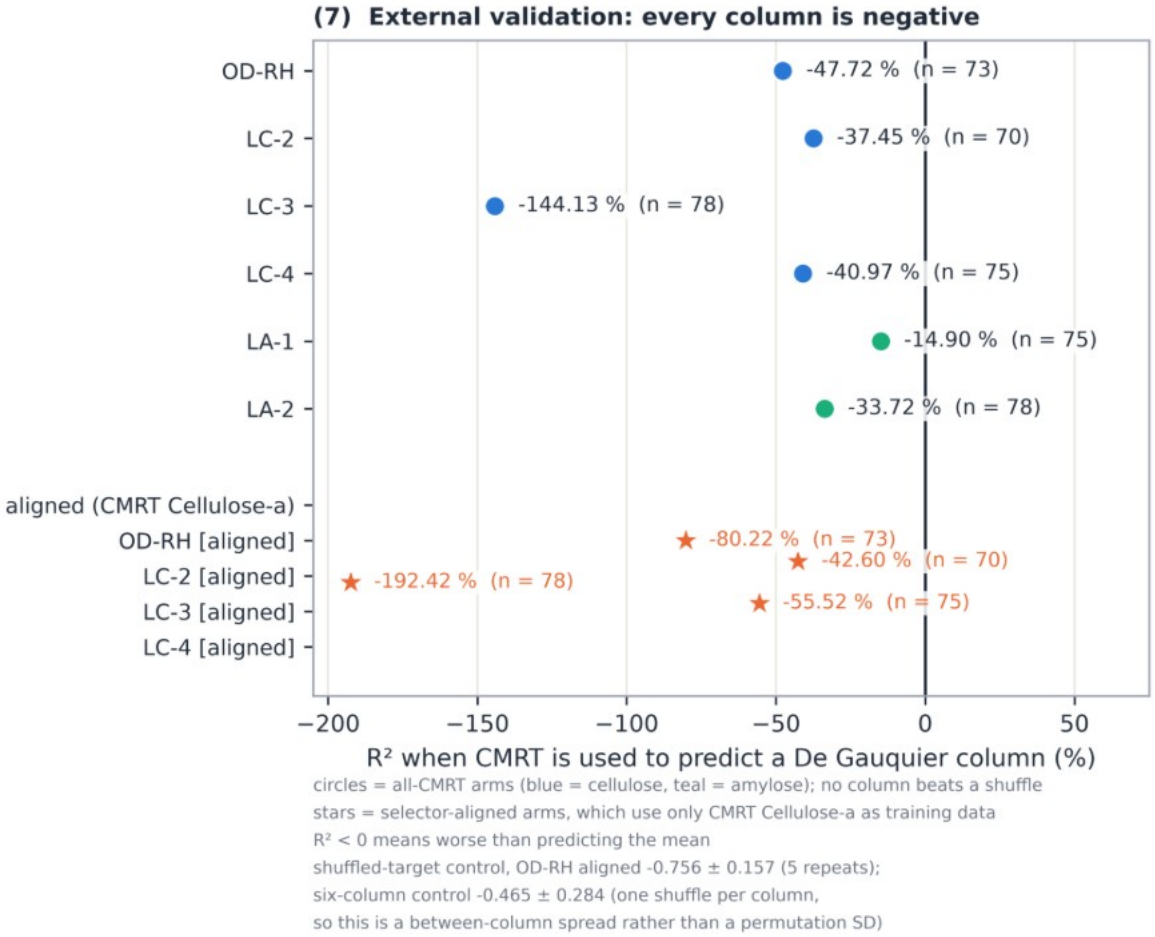


Figure 7 External validation: every De Gauquier column is negative. R^2 when CMRT is used for training and each De Gauquier column (70–78 non-overlapping measurements per column) is predicted; circles are the all-CMRT arms, colored by selector family. The lower asterisk block shows the selector-aligned arms, trained only on CMRT's Cellulose-a family: OD-RH is −80.22 %, and the aligned LC-2, LC-3 and LC-4 are −42.60 %, −192.42 % and −55.52 % respectively, all worse than the corresponding all-CMRT versions (−37.45 %, −144.13 %, −40.97 %), because the training set is four times smaller. Shuffled-target controls are not plotted per column: the aligned OD-RH arm gives −0.756 ± 0.157 over five repeats, so the observed −0.802 is no better than its own control, and both are below the constant-mean predictor ($R^2 = 0$). The six-column control uses only one shuffle per column, so its ±0.284 spread is a between-column spread rather than a permutation standard deviation.

3.7 Quantum-chemical chirality descriptors add nothing

The starting point of this paper is that enantioselectivity requires a chirality descriptor — a pseudoscalar that changes sign between the two enantiomers — and that force-field pseudoscalars should be replaced by quantum-chemical ones. The pseudoscalar construction itself is established work: atom-centered scalar triple products from force-field geometries have already been used for elution-order prediction [6], and the descriptors used in [2] are also scalar triple products of vectors. Our contribution is not the construction but

its systematic falsification at scale: we build three generations of quantum-chemical variants and test them under a cross-study protocol with shuffled-label controls, testing each generation by downstream ablation on four splits (Table 6), and always reporting the fold-to-fold standard deviation of R^2 alongside any ΔR^2.

Table 6 Downstream ablation of the three generations of quantum-chemical chirality descriptors

Arm	S2 grouped	S2b leave-a-batch	S3 leave-column	S4 leave-scaffold
L0 (2D baseline)	0.2759	−0.1088	−0.0711	0.0637
L1a-FF, even magnitude	0.3030	−0.0664	−0.0383	0.1132
L1a-FF, signed	0.3176	−0.0765	−0.0329	0.1277
L1-CF, signed	0.3426	−0.0473	−0.0204	0.1526
L1-CF, even magnitude	0.3335	−0.0534	−0.0214	0.1422
L1-CF + L1a-FF	0.3541	−0.0432	−0.0263	0.1685
L1b, even magnitude	0.2629	−0.0970	−0.0507	0.0402
L1b, signed	0.2500	−0.0885	−0.0481	0.0541
L1-CF + L1a-FF + L1b signed	0.3469	−0.0437	−0.0151	0.1634
Fold-to-fold SD of R^2 (L0)	0.019	0.060	0.389	0.019
Evaluation sample size	10,363	10,363	10,363	10,363
Evaluation sample size (L1b arms only)	10,185	10,185	10,185	10,185
L0 baseline within the L1b subset	0.2721	−0.1106	−0.0581	0.0584

Note: all arms share the same model and the same splits; Δ is measured against the L0 baseline within each descriptor-coverage subset, and should be read together with the rows giving the fold-to-fold standard deviation and the evaluation sample size.

Table 6 mixes two evaluation samples: the L1a-FF and L1-CF arms are evaluated on their joint coverage (n = 10,363), while the L1b arms use the L1b coverage (n = 10,185), so the L0 baseline differs accordingly, being 0.2759 and 0.2721 on S2; rows on opposite sides of that boundary are not directly comparable. The L1-CF design beats L1a-FF on all four splits (Δ = +0.025, +0.029, +0.012 and +0.025 in turn). Against the L0 fold-to-fold standard deviations of 0.019, 0.060, 0.389 and 0.019, Δ exceeds the standard deviation by about 1.3 times on S2 and S4; on S2b the value 0.029 is smaller than 0.060, and on S3 it likewise does not exceed it, so the ordering is consistent but not established. The materiality threshold is taken as $\Delta R^2 = 0.015$, about 0.8 times the L0 fold-to-fold standard deviation on the grouped split; this threshold was not preregistered but fixed after observing that standard deviation, and it is also smaller than the fold-to-fold standard deviations of 0.019 on S2 and S4, so Δ is reported against both. The chirality-specific contrast (signed columns minus even-magnitude columns) amounts to −0.0101 to +0.0146 across both generations and four splits, in absolute value 0.0010 to 0.0146, and for L1-CF alone 0.0010 to 0.0103, everywhere below the threshold; adding L1b makes three of the four splits worse.

The decisive test is the least model-dependent one, the zero-order correlation scan: the per-column |r| against the target, with no model involved. The signed (chirality-bearing) columns do not carry more information than the unsigned (achiral) magnitudes: for L1-CF the median |r| is 0.0352 for the even-magnitude columns and 0.0169 for the signed ones; for L1b it is 0.0315 and 0.0107; for L1a-FF it is 0.0141

and 0.0189. Among the signed columns, the proportions with |r| greater than 0.05 are 2 % (1 of 61), 0 % (0 of 7) and 6 % (3 of 48) respectively. For two of the three generations the signed columns are weaker, and for L1a-FF the two are comparable (0.0189 and 0.0141, both at the noise level); in no case does the chirality-bearing part carry information that the achiral magnitude does not. The scan pools rows across studies, so the between-paper structure is not removed — it is model-free but not grouping-free; the 196 columns are not independent, so we report only the largest single-column |r| (0.079, that is 0.6 % of the variance) without making a significance claim for any column.

This pattern has a mechanistic explanation that was verified rather than assumed, and it marks a limitation of the source data as much as of the descriptors. A pseudoscalar is the product of a sign factor and an even quantity; in CMRT the CIP label of the reference enantiomer is constant by construction (S for 10,516 of the 10,525 rows), so the sign factor has almost no variation and the pseudoscalar is approximately proportional to its unsigned magnitude. The information content of the signed columns therefore cannot be tested on this dataset at all, and the conclusion is restricted to two points: the unsigned components carry no more downstream signal than the magnitudes, and no generation improves cross-study prediction. Whether chirality descriptors would help on a dataset with properly varied enantiomer assignments is not answered here, and could not be answered with the existing labels; this strengthens the data-collection argument (Section 4.4): recording which enantiomer of a pair is the reference is a prerequisite for testing chirality descriptors.

Finally, there are two boundaries. The gain of L1-CF over L1a-FF is real and reproducible (4/4 splits), so descriptor design itself does matter, but it is not the chirality dimension that is responsible; 149 molecules (1.5 %) admit no stereocenter description at all, because the construction requires a four-vertex stereocenter and these molecules carry nitrogen or sulfur stereocenters with lone pairs (sulfoxides, chiral amines), a non-random exclusion of a chemically important class. This is declared in Section 4.5 as well; it is a limitation of the descriptor construction, not a claim that such compounds are unpredictable.

4 Discussion

4.1 Three failures, one structural explanation

Three interventions that are usually considered independent all failed to improve out-of-study prediction: enriching the description on the molecular side, that is, introducing quantum-chemical chirality descriptors (Section 3.7); adding information on the condition side or adapting to the target paper (Section 3.2); and allowing a single oracle affine recalibration of the predictions (Section 3.1). The three can touch only the molecular main effect, the condition main effect and the scale, respectively, whereas the decomposition of Section 3.5 shows that the interaction term plus noise accounts for 76.95 % of the variance in the only dataset in which this interaction can be measured directly. These failures are therefore not three independent negative results but one and the same structural result observed three times; its precise form lies in the interaction term being unidentifiable when each (molecule, condition) cell holds only one observation, rather than in no model class being able to represent an interaction.

This explanation also reframes the earlier finding that repeated measurements are scarce. Within-paper replication is sparse rather than absent: the table contains 609 within-paper replicate groups, and 3.8 % of the pairs report a non-zero relative standard deviation of the retention factor k (up to 0.95), but the median group holds only 2 records, so over most of the table the interaction term and noise remain inseparable. This is precisely the value of repeated measurement: only when the same (molecule, condition) cell is measured more than once can the interaction term be separated from noise. Accordingly, the 23.05 % main-effects value is an upper bound to what an additive description of the label can reach, and how far below it the truly achievable value lies is currently unknown. Adding more molecules does not help with this: more molecules increase the number of cells, not the information per cell; this is consistent with the learning curve of Section 3.3 remaining flat from 1 to 250 training papers.

4.2 What "non-transferable" means and where its boundary lies

It is easy to over-read the above results as "chirality cannot be modeled", but that is not what this paper finds. Within a single paper, molecules unseen by that paper can be predicted with a median per-paper correlation of +0.185, with 80 % of papers positive, and this result remains stable across five random seeds. This value was reached under the condition that the within-laboratory training half holds only about 15 rows; it is the level attainable at that sample size, not an upper bound. The signal does exist; what fails is carrying it to another laboratory: the median across-laboratory per-paper r is only +0.0009 (Section 3.3).

Nor is the failure a matter of scale. If papers differed only by a systematic offset—for example a different column batch or a different calibration—a single additive correction should recover the signal; the oracle experiments of Section 3.1 show exactly how much such a correction can buy, and that it is not enough. Under true leave-one-paper-out, allowing the model to use the mean and scale of each held-out paper itself (one oracle affine recalibration) brings R^2 to 0.439, but this number should not be read as predictive ability. When the within-fold ranking ability is zero, the affine-corrected R^2 equals exactly the between-group variance share; genuine within-paper ranking contributes only +0.06 relative to it. The model-free analysis of Section 3.4 shows the same thing: within one laboratory, changing only the mobile phase largely preserves the ranking ($r = +0.560$), whereas changing both the selector and the mobile phase leaves a median correlation of only +0.084. The selector-only and both-changed arms (30 pairs each) cannot be resolved from each other (paired bootstrap difference +0.061, 95 % confidence interval [−0.059, +0.190]), and the selector's main effect on the target accounts for only about 1 % of the variance (η^2(column) is 1.0 %), with the condition term at 1.4 %; what can be resolved is therefore the mobile-phase-change arm. A small mean shift and an almost complete loss of ranking can coexist, and it is the latter that a predictive model must survive.

The selector result also has a practical corollary that runs counter to intuition: aligning the external validation to the same selector chemistry does not help transfer. Predicting the OD-RH column from CMRT's cellulose tris(3,5-dimethylphenylcarbamate) data gives $R^2 = -0.802$, worse than the −0.477 obtained from the whole of CMRT, for the sole reason that the aligned training set available is four times smaller. Selector alignment reduces one kind of shift while also shrinking the training distribution; when the training distribution itself becomes the limiting factor, this cost exceeds the benefit of alignment. This is

where the boundary of the present conclusions can be drawn. We are speaking of a failure of cross-laboratory transfer, not of chirality itself being impossible to model; of the interaction term being unidentifiable under this data structure, not of no model class being able to learn an interaction. Both qualifications point in the same direction: the bottleneck for transferability lies on the data side, and improvements on the model and descriptor side cannot cross it on their own.

4.3 Recommendations on how accuracy is reported

The first result of this paper is methodological, and it is the one with the most direct practical consequence. On the same underlying data, moving from a grouped split to a true leave-one-paper-out split costs about 19 pp, whereas the residual predictive ability above the majority-class baseline is only +2.2 pp. This cost is not an incidental property of some model but a direct consequence of changing the protocol under the same model and the same feature set. Since the same data can plausibly support reported accuracies spanning this entire interval—the specific value depending on the splitting scheme—chiral chromatography papers should at least supply three reporting items. The role of these two numbers is not to evaluate any individual study but to let readers distinguish how much of a reported accuracy comes from model ability and how much from the splitting scheme.

First, a genuine leave-one-paper-out split: one paper held out per fold, rather than a grouped split in which several papers are divided into groups and held out as whole groups, with the number of held-out papers per fold stated explicitly. Second, a per-study majority-class baseline for elution order: class balance differs enormously between studies, and on the present data the oracle per-study majority-class baseline reaches about 76.6 %, a value in the same range as the accuracies reported in the literature; it is given here to provide an argument for the reporting norm that the baseline should be reported, not as a claim about any specific published result. Finally, any oracle-corrected metric should be reported alongside the uncorrected one, together with the grouping variance share η^2. As shown in Section 3.1, even when the model's ranking ability is zero, the affine-corrected R^2 is bounded below by η^2; an oracle-corrected number without η^2 is uninterpretable.

4.4 What the field should measure instead

The constructive counterpart to the ceiling above is a short list, detailed in Supplementary Information S1. It is deliberately not "collect more data", because the learning curve of this paper shows that doing so does not work. The list has three parts: recording the missing metadata fields, changing the measurement structure, and recording negative results. On the metadata side, several missing fields should be recorded. CMRT records only the i-PrOH fraction and the flow rate, whereas buffer identity and concentration, pH, column temperature, gradient program, additives, a measured (rather than estimated) dead time and the column batch number are all first-order determinants of enantioselectivity and are currently all absorbed into the unexplained study-level variance. Recording these fields costs nothing in instrumentation and requires no additional experimental design; it is only necessary to reserve fields for them in the data tables.

On the measurement structure, a design different from the existing data should be adopted. At least one molecule × condition grid of about 30–50 molecules systematically spanning ≥4 conditions would allow

the interaction term to be estimated rather than confounded—which is precisely why the De Gauquier grid is informative here despite containing only 43 compounds. The role of the grid is not to increase the sample size but to bring the behavior of the same molecule under different conditions into one comparable structure. Replication with ≥3 measurements per cell, one of them completed on a different day, would separate interaction from noise and would allow the field to answer the question that the present analysis cannot: how much of the apparent "unpredictability" is merely measurement fluctuation.

On negative results, failed separations should be recorded. Only 0.16 % of the CMRT pairs satisfy $\alpha \equiv 1$, against 49–72 % (58 % overall) of the non-overlapping samples in our external dataset, showing that this dataset retains almost exclusively successful separations. This difference has two implications: part of the external-validation failure comes from the task definition rather than from the model itself; and a set composed only of successful separations cannot teach a model to judge whether a given analyte can be separated at all. Screening CSPs is precisely the task in which "this column will not separate this analyte" is the most valuable, so the absence of failed attempts is a systematic blind spot.

4.5 Limitations

Beyond the individual result-level limitations listed in Sections 3.2–3.7, three apply to the study as a whole. First, the variance decomposition underlying Section 3.5 comes from a single dataset, namely 43 molecules in one laboratory; it demonstrates a mechanism rather than a universal constant, and in particular 23.05 % should not be read as a bound for CMRT. Second, CMRT's replication is too sparse to separate predictability from target noise: 609 within-paper replicate groups, 3.8 % of pairs carrying a non-zero relative standard deviation of the retention factor k, a median group of two records and no cell with more than eight records, so the present paper cannot achieve this separation.

Third, the external dataset overlaps CMRT in four analytes (these records have been removed), and the mobile-phase modes of the two differ, so the external-validation failure is a composite of laboratory and condition effects that we cannot fully disentangle. Taken together, the three limitations define the scope of the claims made in this paper: the conclusions are mechanistic and directional, and their value lies in explaining where the failure occurs rather than in giving a specific estimate on some dataset. In addition, the descriptor-construction limitation described in Section 3.7 applies equally to the interpretation of the chirality descriptors throughout the paper. The four-vertex stereocenter construction excludes, non-randomly, compounds whose stereocenters contain lone-pair nitrogen or sulfur (sulfoxides, chiral amines); this is a limitation of the descriptor construction and does not mean that such compounds are unpredictable.

5 Conclusions

On the largest public chiral chromatography dataset, cross-study prediction of enantioselectivity is not distinguishable from chance: the elution-order accuracy is 52.85 % (95 % CI 50.63–55.15, from a bootstrap clustered by publication), which looks 2.18 pp higher than the 50.67 % global majority-class baseline, but the within-paper permutation null itself already reaches 52.33 % ± 0.58 pp, the observed value is not distinguishable from it ($z \approx 0.89$, empirical $p \approx 0.13$), and it is far below the level reported in this field; the

per-study oracle majority-class baseline is 76.55 %, in the same range as the 76–96 % elution-order accuracies reported in the literature. Two baselines need to be distinguished: the paper uses the global majority class as the reference because the class balance of a new study cannot be known in advance; the per-study oracle baseline serves only to quantify how much of the apparent difficulty comes from study-level class imbalance. The shuffled-label control used only three permutations, and the row-level control band is itself too narrow to establish significance under study-level clustering; we therefore built the within-paper permutation null directly on the main protocol (S2c, 530 folds, 30 within-paper permutations, retraining all folds each time), and checked it at the same time on the cheaper whole-paper holdout protocol (S2b, 10 folds, 200 permutations per scheme).

Within-study permutation, which preserves each study's own class balance and destroys the label–feature pairing, is the conservative and relevant null. On S2b its null accuracy is 51.79 % ± 0.63 pp (95 % range 50.71–52.88 %), against the observed 53.94 %, $z \approx 3.4$; block permutation of whole-study label vectors additionally removes the study-level class-balance structure that the model can legitimately exploit, giving 50.54 % ± 0.55 pp ($z \approx 6.2$). On the main protocol S2c, however (530 folds, 30 within-paper permutations, retraining all folds each time), the null accuracy is 52.33 % ± 0.58 pp (95 % range 51.36–53.36 %), and the observed 52.85 % is not distinguishable from it ($z \approx 0.89$, empirical $p \approx 0.13$). The +2 pp residual that appeared detectable on S2b therefore does not reproduce on the main protocol; the more accurate description is "not distinguishable from chance", and the +2.18 pp gain of S2c over the global majority class comes mainly from the study-level class structure that the within-paper null itself already reaches, and does not constitute evidence of molecular-side skill. This is consistent with the across-paper per-paper correlation $r = +0.0009$. The failure is not in model capacity or descriptor quality. The molecule-level signal is firmly present within a study, whereas in the only fully crossed grid available 76.95 % of the variance lies in a molecule × condition interaction term that is not identifiable under this paper's data structure. This means that progress depends on the field's data-collection design, including supplementary condition metadata, at least one systematic molecule × condition grid, repeated measurements and the recording of failed separations, rather than on more compounds or more expensive descriptors. We release the curated dataset, the evaluation protocols, the decomposition scripts and the figures to support this reorientation.

Declarations

Ethics approval and consent to participate

This study does not involve human or animal subjects and uses only publicly available, previously published data; ethics approval and consent to participate are therefore not applicable.

Consent for publication

Not applicable.

Data and materials availability

The source dataset CMRT and the two datasets used for external validation have all been made public previously [11, 16, 17]; the CMCDS dataset used for the L1b descriptors of Section 2.5 is likewise publicly available [14]. This paper provides the curated data tables (10,525 enantiomer pairs), the indices of the seven evaluation splits, the feature matrices, and the analysis scripts that regenerate every number and figure in this paper, all provided as Supplementary Data (repository: https://github.com/CesiumAuride526/chiral-ml-paper1; an archival Zenodo DOI will be added when available). No new experimental data were generated in this study.

The Supplementary Data tables used in the above analyses are released with this paper. They cover the dual-scheme bootstrap, the power-curve criterion and the η^2 samples, the elution-order accuracy of additive rules, the compound-clustered intervals for the condition-pair comparisons of Section 3.4, and the permutation null distributions evaluated on the S2b and S2c protocols.

Competing interests

The author declares no competing interests.

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Author contributions

The author conceived and designed the study, curated the dataset, developed the analysis code, performed all analyses and validations, interpreted the results, and wrote the manuscript. The author read and approved the final manuscript.

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Use of AI-assisted tools

An AI coding assistant was used to draft and refactor the analysis scripts, to run the reproducible checks reported in this paper, and to assist with language editing. All scientific claims, numerical results and interpretations were reviewed and verified by the author; the AI tool is not an author and takes no responsibility for the content.

Supplementary material

The Supplementary Information contains eleven parts (S1–S11), several of which are cited in this paper. S1 gives the metadata fields and measurement structures that should be recorded, namely seven missing

condition fields, three measurement-structure requirements and a ready-to-use collection template; S4 and S7-A give the seven splits with their fold sizes and the three fold-count conventions. S7-C and S7-D give the per-fold failure-mode decomposition and the S2c out-of-fold summary; S7-K gives the elution-order accuracy of additive rules on the De Gauquier grid (oracle two-way 72.09 %, leave-one-molecule 76.28 %, against a 58.60 % majority-class baseline); S7-L gives the PE25 external-validation table. S8 and S8.4 give the negative controls, the power analysis and the permutation nulls on the S2b protocol (within-study null 51.79 % ± 0.63 pp, against the observed 53.94 %), and S10 gives the complete ten-item disclosure list. S11 reports the five additional analyses added in this version: the within-paper molecular-versus-condition attribution (H1), the paper-clustered confidence intervals (H3), the leave-one-selector-family extrapolation (S3′), the per-column leave-one-paper-out evidence (H7), and the 530-fold within-paper permutation null on S2c (H2; null 52.33 % ± 0.58 pp, $z \approx 0.89$, empirical $p \approx 0.13$).

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